MineMind

JV Inkai: acid delivery timing and wellfield commissioning

Which new block groups start later when acid arrives late?

A source-informed, representative digital twin on OpenEgiz.
18 twins. One adjustable scenario. Complete reset.

Synthetic block groups and schedules. Assumed commissioning response.
No site calibration, optimizer or actual operating forecast.

Connected platform and mine flows

Model publisher -> MQTT -> Eclipse Ditto -> spatial interface

The UI reads all operating states and projections back from Ditto.
Missing twins or mixed revisions produce an error, not substitute model data.

Commissioning acid

Separate area stocks -> whole-dose reservation -> acidification -> commissioning

Uranium-bearing solution and product

Block extraction -> area IX -> recovered-U queue -> shared MPP drying

Existing/sustaining acid supply is protected outside this model. Satellite transfer is aggregated.

Evidence and explicit assumptions

Public facts

MPP / Sat-1 / Sat-2 IX: 2.7 / 6.3 / 4.5 Mlb U3O8/year.
Drying: 8.3 Mlb/year. Plant recovery: approximately 98%.

Synthetic exercise and assumptions

Nine group identities, positions, stocks, rates and schedules.
Whole commissioning doses: 100-160 t per new group.
Seven-day acidification, seven-day ramp, fixed 100 mg U/L.
Annual capacities converted to uniform daily model limits.
One drying-capacity day of end-of-day WIP: an assumed policy.

The operator reported new-block acidification delays despite 106% nine-month acid fulfilment. This supports the timing mechanism, not these coefficients or a 28-day reconstruction.

Sources: Cameco 2024 Inkai report, printed pp.80-82; Kazatomprom 3Q24 update; PARAMETERS.md

Accountable inventories and constraints

Acid stock = initial stock + deliveries - reserved doses

Extracted U = dried product U + queue U + unrecovered U

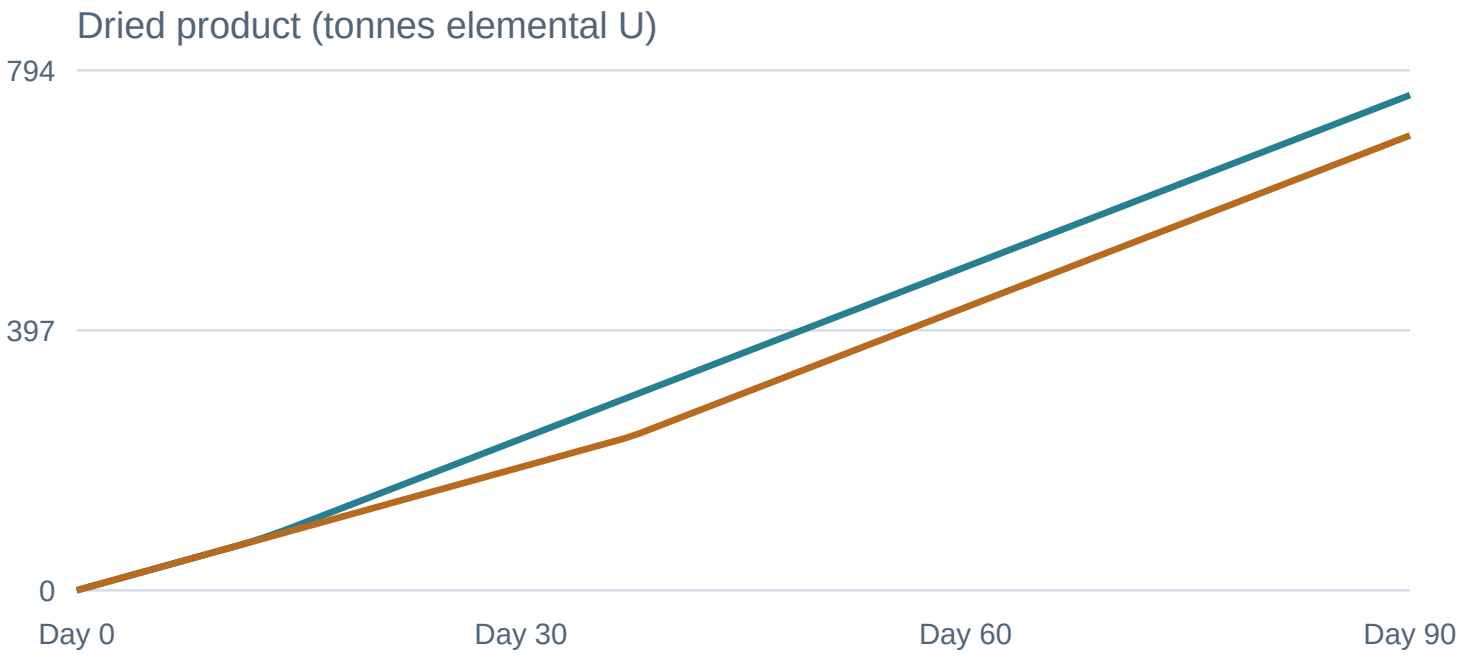
Initial group U = remaining U + cumulative extraction

Area IX limits plus drying/WIP headroom pace extraction.
End-of-day recovered-U WIP stays at or below 8.75 t U.
Capacity-withheld potential extraction stays underground.

The WIP target is not a measured tank size. Conservation is internal consistency, not site validation.

Simulated delivery-delay comparison

First arrivals: day 3 -> day 31 (+28 days). Second arrivals stay day 60.



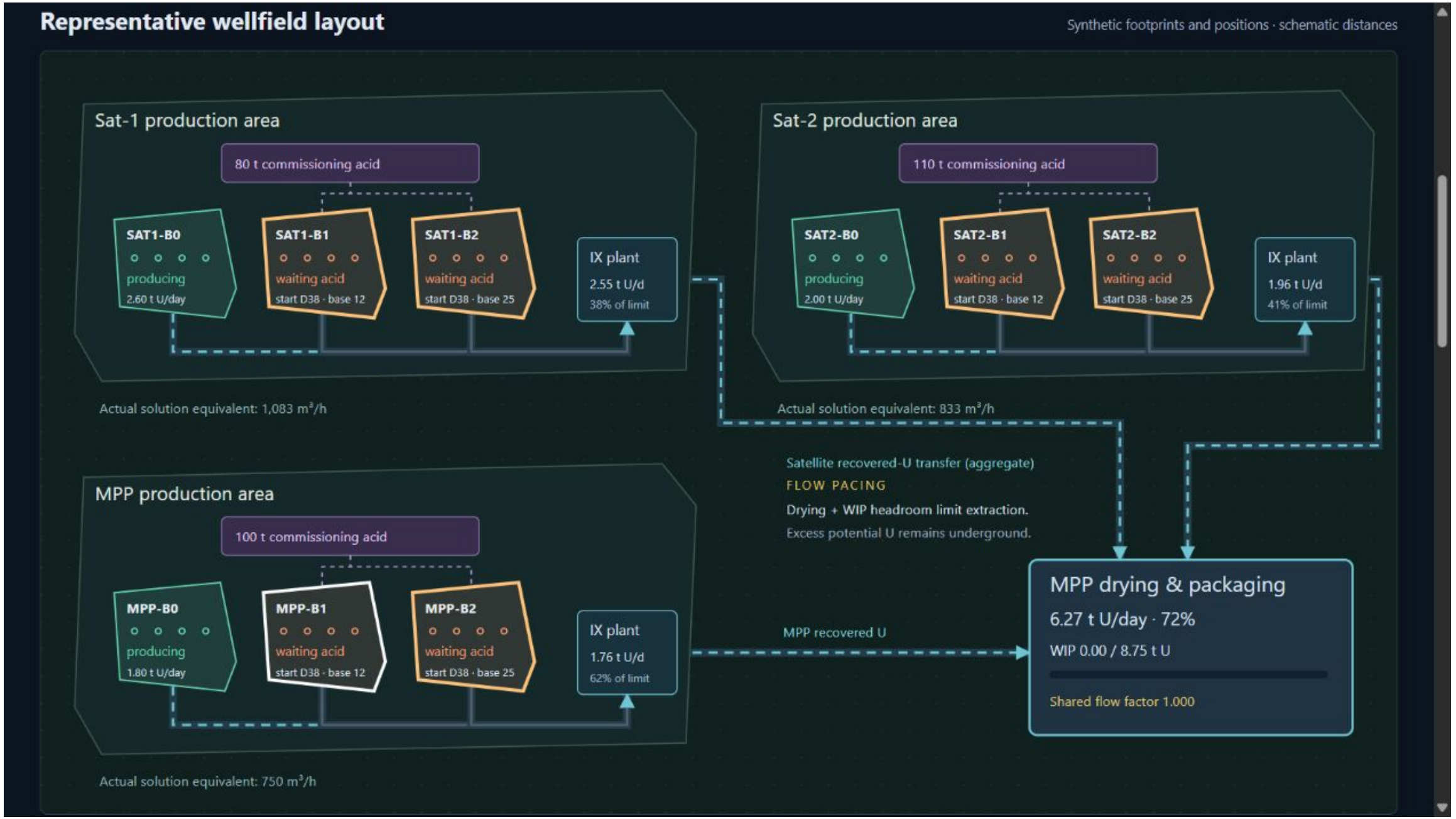
Baseline

28-day delay

Day 30: 231.30 vs 188.16 t U.
Day 90 difference:
-61.66 t U.

New B1 starts: D12 -> D38.
Same total acid by day 90.

A model outcome, not an Inkai loss estimate. Different assumptions or bottlenecks can change the difference.


Representative wellfield layout
Synthetic footprints and positions · schematic distances
Sat-1 production area
80 t commissioning acid
SAT1-B0
producing
2.60 t U/day
SAT1-B1
waiting acid
start D38 · base 12
SAT1-B2
waiting acid
start D38 · base 25
IX plant
2.55 t U/d
38% of limit
Actual solution equivalent: 1,083 m³/h
Sat-2 production area
110 t commissioning acid
SAT2-B0
producing
2.00 t U/day
SAT2-B1
waiting acid
start D38 · base 12
SAT2-B2
waiting acid
start D38 · base 25
IX plant
1.96 t U/d
41% of limit
Actual solution equivalent: 833 m³/h
Satellite recovered-U transfer (aggregate)
FLOW PACING
Drying + WIP headroom limit extraction.
Excess potential U remains underground.
MPP production area
100 t commissioning acid
MPP-B0
producing
1.80 t U/day
MPP-B1
waiting acid
start D38 · base 12
MPP-B2
waiting acid
start D38 · base 25
IX plant
1.76 t U/d
62% of limit
Actual solution equivalent: 750 m³/h
MPP recovered U
MPP drying & packaging
6.27 t U/day · 72%
WIP 0.00 / 8.75 t U
Shared flow factor 1.000

How robust is the conclusion?

27 cases: dose x0.75 / x1 / x1.25; acidification 3 / 7 / 14 days; WIP 0 / 1 / 7 days.

Every delayed case uses its own no-delay baseline and the same accounting model.

Day-90 product difference (t U), at a one-day WIP target

Dose multiplier	3 days	7 days	14 days
0.75x	0.00	0.00	0.00
1.00x	-61.66	-61.66	-61.66
1.25x	-61.66	-61.66	-61.66

Across all cases: 3-6 new groups start late.

A product shortfall is NOT robust across these assumptions.

Illustrative perturbations, not measured bounds or confidence intervals. Lower assumed doses let B1 start from initial stock; capacity can mask B2 delays. No optimized policy or real loss estimate.

Verification and reproduction

29 unit tests: 26 model + 3 mocked read-contract tests.
12 live application groups + 23 platform smoke checks.
Browser observations: spatial states, sensitivity, controls, reset.

Evidence boundary

Current runtime: fresh isolated full-stack WSL2 startup in 83.60 seconds.
New credentials and empty volumes; cached images and build layers.
Cold downloads and exact judge KVM remain unverified.

Start on Ubuntu with Docker installed:
sudo apt-get update && sudo apt-get install -y make
make inkai-test && make inkai && make inkai-smoke

Open localhost:8088. Baseline day 30 -> Delay 28 days -> inspect days 38 / 90.
Review sensitivity. Reset - day 0 restores every initial physical state.